

If $\{\varepsilon_j, j \geq 1\}$ are centered two-sided Pareto distributed with parameter α , it follows by some basic calculations that

$$f_X(x) = \frac{1}{x} \int_{\mu}^{\infty} h_{\alpha}(x, u) du 1_{\{x > 0\}} - \frac{1}{x} \int_1^{\mu} h_{\alpha}(x, u) du 1_{\{x < 0\}},$$

$$J_1(\Psi_x \circ \sigma) = - \int_{\mu}^{\infty} h_{\alpha}(x, u) du 1_{\{x > 0\}} + \int_1^{\mu} h_{\alpha}(x, u) du 1_{\{x < 0\}},$$

where μ is the expected value of $\tilde{\varepsilon}_1$, i.e. $\mu := \frac{\alpha}{\alpha-1}$, and

$$h_{\alpha}(x, u) = \alpha u^{-\alpha-1} \varphi \left(\frac{1}{2} [\log((u - \mu)^2) - \log(x^2)] \right) 1_{\{u > 1\}}.$$

As a result, we have

$$\int J_1(\Psi_x \circ \sigma) dF_{\psi(X_1)}(x) = \int_0^{\infty} \frac{1}{x} \left\{ \left(\int_1^{\mu} h_{\alpha}(x, u) du \right)^2 - \left(\int_{\mu}^{\infty} h_{\alpha}(x, u) du \right)^2 \right\} dx.$$

Since our theoretical results do not provide a non-degenerate asymptotic distribution for the Wilcoxon test statistics, we only consider CUSUM and self-normalized CUSUM test for the first scenario (when the $\varepsilon_j, j \geq 1$, are standard normally distributed).

		CUSUM				self-norm. CUSUM			
H	n	$h = 0$	$h = 0.5$	$h = 1$	$h = 2$	$h = 0$	$h = 0.5$	$h = 1$	$h = 2$
0.6	500	0.046	0.384	0.958	1.000	0.043	0.372	0.853	0.998
	1000	0.045	0.728	1.000	1.000	0.047	0.627	0.975	1.000
	2000	0.043	0.958	1.000	1.000	0.044	0.856	0.998	1.000
0.7	500	0.064	0.392	0.962	1.000	0.044	0.395	0.858	0.995
	1000	0.056	0.736	0.999	1.000	0.044	0.629	0.974	1.000
	2000	0.050	0.964	1.000	1.000	0.046	0.869	0.997	1.000
0.8	500	0.074	0.369	0.969	1.000	0.039	0.462	0.860	0.993
	1000	0.074	0.734	1.000	1.000	0.044	0.670	0.963	0.999
	2000	0.072	0.966	1.000	1.000	0.048	0.851	0.996	1.000
0.9	500	0.095	0.311	0.977	1.000	0.045	0.583	0.875	0.980
	1000	0.101	0.739	0.998	1.000	0.045	0.736	0.946	0.996
	2000	0.101	0.979	1.000	1.000	0.043	0.860	0.987	1.000

Table 1: Rejection rates of the CUSUM tests for LMSV time series (standard normal $\varepsilon_j, j \geq 1$) of length n with Hurst parameter H and a shift in the mean of height h after a proportion $\tau = 0.5$. The calculations are based on 5,000 simulation runs.

Note that in this case the size of the non-self-normalized CUSUM change point test